

MitoVis: A Unified Visual Analytics System for End-to-End Neuronal Mitochondria Analysis

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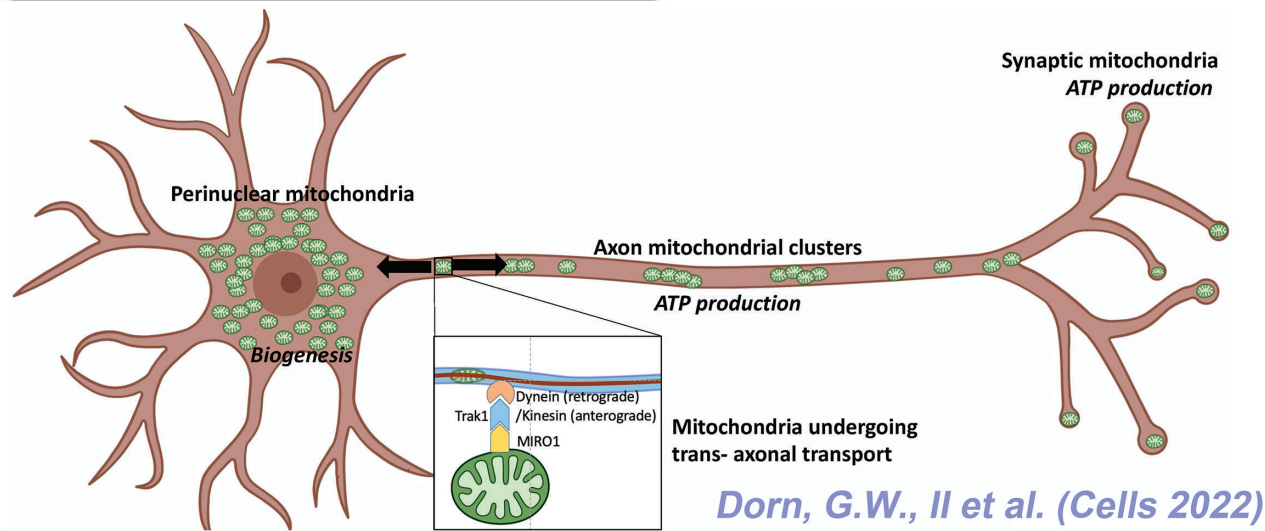
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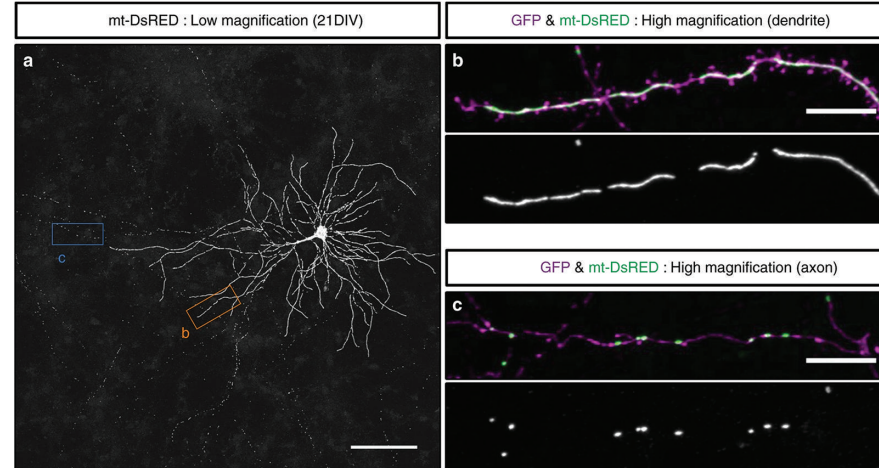
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Mitochondria Analysis

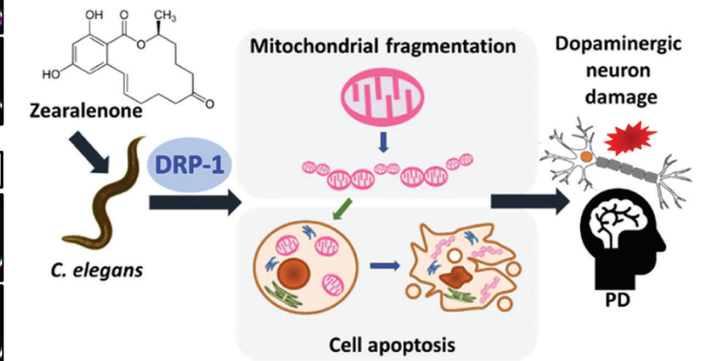


Mitochondria is a type of the intracellular organelles displaying compartmentalized features in the dendrites and axons of the neuron.

Lewis & Kwon et al. (Nature comm. 2018)



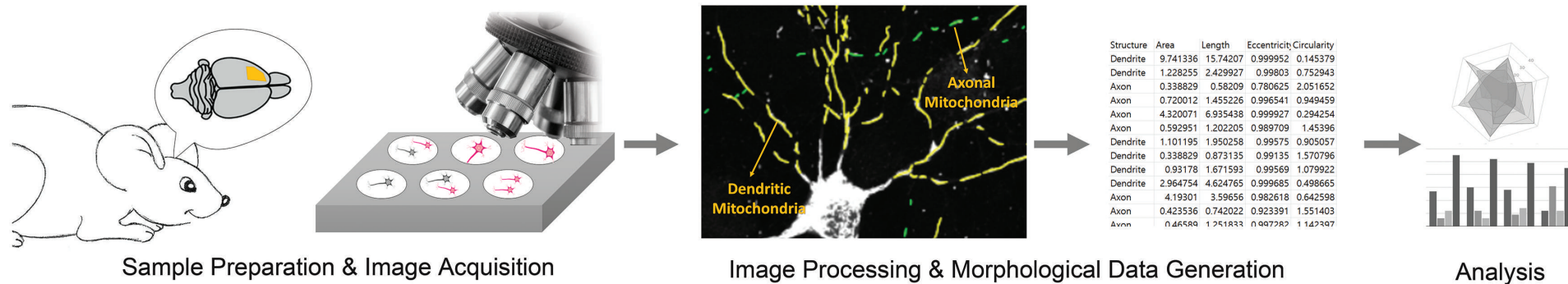
Wei et al. (J. Agric. Food Chem. 2021)



The difference in mitochondrial morphology is known to be associated with the function of each compartment such as neurotransmitter release and dendritic protein synthesis via calcium clearance and ATP generation. Moreover, fragmentation of dendritic mitochondria has been observed in neurodegenerative disease models such as Alzheimer's and Parkinson's diseases.

Problem

Conventional Analysis Flow

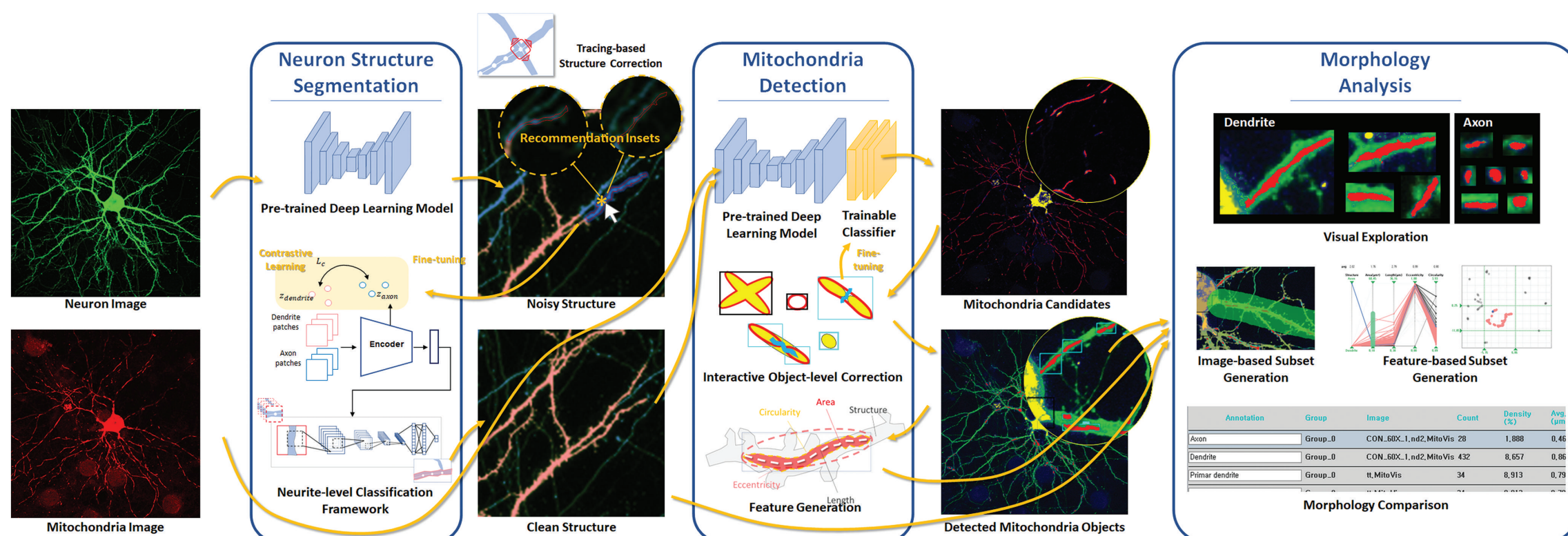


- ✓ Conventional analysis is mainly based on **manual annotations and classification**, which is time consuming and labor intensive.
- ✓ Deep learning-based algorithms **includes many errors**, demanding additional manual proofreading.
- ✓ There is **no unified system** providing all the necessary functions for data processing and visualization.

MitoVis

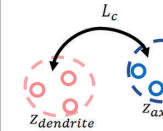
Contribution 1. A novel visual analysis system specifically designed for end-to-end neuronal mitochondria morphology analysis on light (confocal) microscopy data.

Contribution 2. A novel visualization-guided data suggestion scheme combined with a contrastive learning and tracing-based neurite-level classification for fast and accurate fine-tuning.



Neurite-level interaction

A neuron is decomposed into neurites using a neuron-tracing method. The user interaction and classification are performed on the neurite level. This enables fast interaction and noise robust classification.



Contrastive Learning

Contrastive loss improves the feature distinction capability. With contrastive loss, the model will be able to learn the global context of data from many different patches.

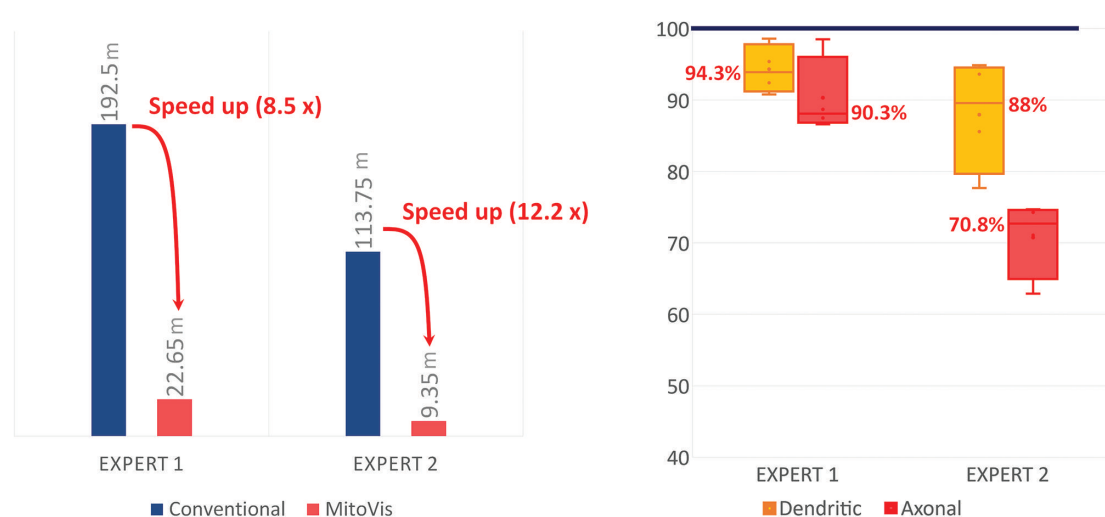


Visual guide

The visual guides provide spatial clues to users, enabling them to quickly identify the locations of errors in the labels of the structure and the mitochondria in order to correct them.

Evaluation

Performance comparison between conventional analysis and MitoVis.



The analysis time for the conventional analysis method and MitoVis.

The average analysis score of MitoVis.
* Average analysis score: The average relative length compared to the average length of the manual measurement

Conclusion and Future Work

MitoVis enables

- rapid analysis** by drastically reducing the time required for data processing through deep learning
- precise analysis** through effective visualization, interaction and interactive learning approaches
- flexible analysis** by allowing the generation of various mitochondria subsets using feature based and image-based subset generation methods

In the future, we plan to

- conduct **In-depth analysis** of disease models and neuronal cell types using MitoVis
- expand the current system for analyzing **3D neuronal images and different modalities** (e.g., electron microscopy)

References

- [1] Dorn, Gerald W., and Xiawei Dang. "Predicting Mitochondrial Dynamic Behavior in Genetically Defined Neurodegenerative Diseases." Cells 11.6 (2022): 1049.
- [2] Lewis, Tommy L., et al. "MFF-dependent mitochondrial fission regulates presynaptic release and axon branching by limiting axonal mitochondria size." Nature communications 9.1 (2018): 1-15.
- [3] Wei, Chia-Cheng, et al. "Zearalenone Induces Dopaminergic Neurodegeneration via DRP-1-Involved Mitochondrial Fragmentation and Apoptosis in a Caenorhabditis elegans Parkinson's Disease Model." Journal of Agricultural and Food Chemistry 69.40 (2021).

Acknowledgement

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